

- 2 The summer squash plant, *Cucurbita pepo*, produces edible fruits that vary in shape. Fig. 2.1 shows the fruits of three different varieties of squash plants.



Fig. 2.1

Fruit shape in squashes is controlled by two genes, **A/a** and **B/b**, that are located on different chromosomes.

- A disc-shaped fruit is produced when both dominant alleles, **A** and **B**, are present.
- A spherical fruit is produced when either allele **A** or allele **B** is present, but not if both **A** and **B** are present.
- A long fruit is produced when both allele **A** and allele **B** are absent.

- (a) (i) Table 2.1 shows the possible genotypes of the Patty pan and Alfresco varieties.

Complete Table 2.1 to show the possible genotypes of the Di Nizza variety.

Table 2.1

variety	possible genotypes			
Patty pan (disc-shaped)	AABB	AaBB	AABb	AaBb
Di Nizza (spherical)				
Alfresco (long)	aabb			

[1]

- (ii) A gardener used pollen from a male flower of Alfresco to pollinate a female flower of Di Nizza. The gardener grew the seeds produced from this cross and found that half the offspring produced spherical fruits and half produced long fruits.

Draw **one** genetic diagram to explain this result.

parent genotypes

gametes

offspring
genotypes

offspring
phenotypes

[4]

- (iii) The offspring show genetic variation with respect to fruit shape alleles.

Name the process that occurred during meiosis in the parents that produced this variation **and** state the stage of meiosis at which it occurred.

process

stage of meiosis [2]

- (b) (i) Genetically modified (GM) summer squash plants with resistance to viral diseases have been grown in the USA since 1995.

Scientists have been concerned that viral resistance genes pass easily from GM squash plants to their wild relative, the Texas gourd, *Cucurbita texana*.

Explain why the possibility of gene flow from GM squash plants to the Texas gourd is a cause of social and environmental concern.

social

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environmental

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[2]

- (ii) A study compared the survival of two different types of hybrids that were formed by cross-pollination between GM virus-resistant squash plants and wild Texas gourd plants:
- virus-resistant hybrids that had inherited the viral resistance gene
 - non-resistant hybrids that had **not** inherited the viral resistance gene.

Viral disease outbreaks caused many of the non-resistant hybrids to die. Leaf-eating beetles then moved in larger numbers to the surviving healthy virus-resistant hybrids. The beetles carried a pathogenic bacterium *Erwinia* which was capable of killing the plants.

Fig. 2.2 compares infection with *Erwinia* in the virus-resistant hybrids and the non-resistant hybrids.

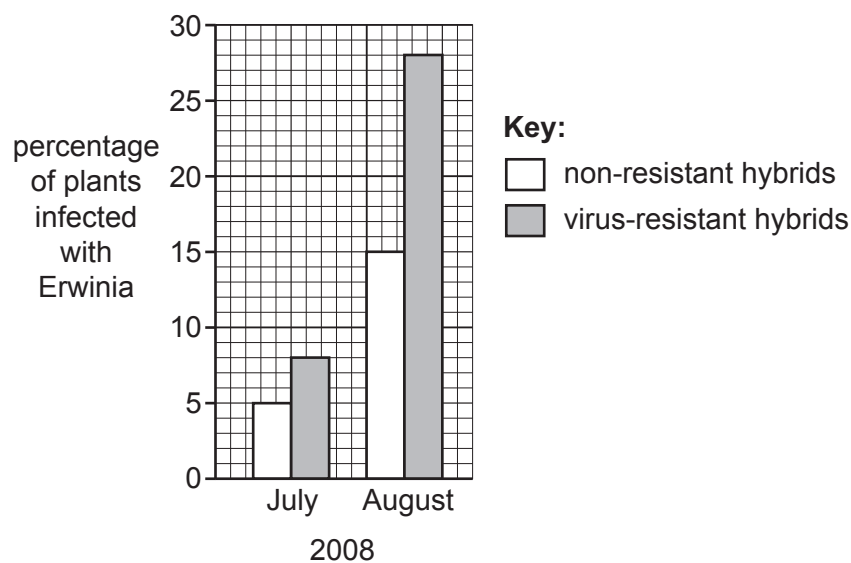


Fig. 2.2

Discuss whether these results provide support for the use of genetically modified organisms (GMOs) in food production.

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- (iii) Suggest why the GM virus-resistant squashes grown by farmers rarely suffer infection by *Erwinia*.

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[Total: 12]